

UMUT-II HYDROPOWER PLANT, TURKEY



Document Prepared By TÜV Rheinland Energie und Umwelt GmbH

Project Title	Umut-II Hydropower Plant, Turkey
Version	1.0

Report Title	2011-09-13 Umut-II 21216157d
Client	Nisan Elektromekanik Enerji San. ve Tic. A.S.
Pages	75
Date of Issue	23 September 2011
Prepared By	TÜV Rheinland Energie und Umwelt GmbH
Contact	Am Grauen Stein, 51105 Cologne, +49 221 806 3553, carbon@tuv.com , www.tuv.com/carbon
Approved By	Norbert Heidelbergmann
Work Carried Out By	Roland Wollenweber, Jürgen Wiesmann

Summary:

Validation of a 24.45 MWe HEPP according to VCS v.3 applying AMS-I.D. "Grid connected renewable electricity generation" v.12.1.in combination with the "Tool to calculate the emission factor for an electricity system", v.02.1.0 and the CDM Additionality Tool, v.5.2. Validation was made to a reasonable level of assurance also applying the requirements outlined in VCS v.3. The project information validated includes the VCS PD and all relevant information and supporting evidence acquired during the validation process. The validation team has come to the conclusion that based on the on-site assessment and all available documentation the GHG assertion is made in accordance with the requirements of the VCS programme, is materially correct and fairly represents the GHG emissions data and information without material discrepancies.

Table of Contents

Introduction	3
1.1 Objective	3
1.2 Scope and Criteria	3
1.3 Level of assurance	4
1.4 Summary Description of the Project	4
2 Validation Process	4
2.1 Method and Criteria	4
2.2 Document Review	5
2.3 Interviews	7
2.4 Site Inspections	8
2.5 Resolution of Any Material Discrepancy	8
3 Validation Findings	8
3.1 Project Design	8
3.2 Application of Methodology	9
3.2.1 Title and Reference	9
3.2.2 Applicability	9
3.2.3 Project Boundary	9
3.2.4 Baseline Scenario	9
3.2.5 Additionality	10
3.2.6 Quantification of GHG Emission Reductions and Removals	11
3.2.7 Methodology Deviations	12
3.2.8 Monitoring Plan	13
3.3 Environmental Impact	13
3.4 Comments by stakeholders	13
4 Validation conclusion	14
5 Appendix A	15

INTRODUCTION

1.1 Objective

The purpose of the Validation Report is to confirm that the Umut-II Hydroelectric power plant project (hereafter the project) and all related project documentation conforms with all rules and requirements of the VCS Standard v.3 and the applied GHG methodology, here ACM0002, version 12.1.

Further the validation shall be made for an initial crediting period of 10 years as being part of 3 x 10 years of crediting period in total. Therefore this is to be considered the first validation process for this project. Upon successful validation the project shall be approved by the VCS program to be included as a registered VCS project generating Verified Carbon Units (VCUs).

1.2 Scope and Criteria

The criteria for validation are the rules and requirements of the VCS Standard V.3, in particular the principles articulated in section 2.4 of the VCS Standard V.3. The VCS-PD applies an approved methodology from the UN Clean Development Mechanism (CDM). Therefore, the methodology ACM0002, version 12.1, the methodological tool “Tool to calculate the emission factor for an electricity system, version 2” and the CDM Additionality Tool, version 5.2 are additional criteria for the project.

The scope of validation is set in section 3 of the VCS Standard V.3. It includes the following key areas:

1. the VCS-PD is complete and fulfils the requirements of section 3.19 of the VCS Standard V.3
2. the chosen methodology ACM0002 is applicable to the project
3. the project boundary is identified in accordance with ACM0002
4. the baseline is chosen in accordance with ACM0002 and properly justified
5. the emission factor for grid-based electricity is established in accordance with the “Tool to calculate the emission factor for an electricity system”.
6. the data and parameters for the quantification of the GHG emission reductions are provided in accordance with ACM0002
7. the monitoring plan establishes a reliable GHG information system for obtaining, recording, compiling and analyzing data for the quantification of GHG emission reductions.
8. the additionality of the project is demonstrated in accordance with the CDM Additionality Tool

1.3 Level of assurance

The Validation Statement is made to a reasonable assurance level, and a materiality threshold of 5% is applied for identification of material omissions as required by VCS v.3 paragraph 5.3.1. item 4).

In order to comply with the requirements of VCS v.3 the validation shall be carried out selecting data and evidence to provide a reasonable level of assurance as stated in the VCS v.3 paragraph 5.3.1. item 1). In general it shall be adhered to the procedures as defined in ISO 14064-3.

1.4 Summary Description of the Project

The project is a run-of-river hydropower project located on the Karakus River, Ordu Province, Turkey. The project consists of multiple diversion weirs and a power station with 3 Francis-type turbines with a gross head of 152 m and a project flow of 18 m³/sec. The weirs have a reservoir with roughly 0.114 km² surface area. The turbines have a capacity of 8.15 MWe each, totalling to 24.45 MW, according to the turbine purchase contract. The estimated net power supply to the grid is 68.74 GWh per year. The generated power will be supplied to the national power grid via a 154 kV transmission line.

The validation team was able to confirm the accuracy of the project description by reviewing maps and technical drawing during the site visit as well as via the on-site inspection.

2 VALIDATION PROCESS

2.1 Method and Criteria

Validation Schedule

Contract award for validation	2011-01-28
Receipt of Draft VCS PD	2011-02-20
Desk Review Findings	2011-04-25
On-site assessment	2011-05-23-27
Validation Protocol	2011-06-01
Draft Validation Report	2011-09-13
Internal Technical Review	2011-09-23
Final Validation Report	2011-09-23

The validation process begins when the PPs submit the VCS PD to the validation team along with additional supporting information, in particular spreadsheet for the calculation of the combined margin grid emission factor and the investment analysis. The validation team then reviews the documents and submits the desk review findings to the PPs. During the on-site visit, the validation team meets with the project company at their offices as well as the project site. The validation team confirms that the actual project conforms to the VCS PD, collects further supporting documents and discusses the desk review findings with the PPs. Following the site visit a validation protocol is shared with the PPs. The protocol

lists all Corrective Action Requests (CARs) and Clarification Requests (CLs) The CAR/CLs are then closed by the PPs who are delivering additional supporting evidence and make necessary amendments to the VCS PD. Once all CARs/CLs are closed the validation team drafts the validation report. It is issued, after successfully passing the Internal Technical Review.

2.2 Document Review

The documents as indicated below have all been reviewed during the validation process.

No	Document	Comment
1	VCS Project Description	Dated 06.09.2011
2	EPDK license	Issued by Energy Market Regulatory Authority (EPDK), dated 01.05.2008
3	Turbine purchase contract	Dated 05.06.2008
4	Feasibility study	Dated February 2009
5	Water Utilization Agreement with DSI	Dated 20.03.2008
6	Amendment to Water Utilization Agreement with DSI	Dated 18.06 2010
7	Forest Permit	Dated 16.09.2008
8	Preliminary Environmental Impact Assessment – Project Introduction File	March 2008
9	EIA Not Required Statement	Dated 12.05.2008
10	Board decision on seeking carbon financing for HEPPs	Board decision of PO members Halil Ataman and Hüseyin Calik to consider carbon financing by majority decision with the meeting held on 06.10.2008
11	Carbon development agreement	Contract between the project developer and the project owner to develop eight HEPP projects as Carbon Projects under GS and VCS. The agreement is made of July 9th 2009.
12	Carbon revenue proposal PD	A carbon project development proposal made by the PD that was initially assigned by PO. The proposal is made of April 24, 2008.
13	VERPA cover page	VERPA cover page of the initial VERPA between the first PD that entered into cooperation with the PO. The VERPA is dated 21 August, 2008

14	Technical project design drawings	AutoCAD drawing documents including the technical design of the dam area and intake structures. The documentation is of such detail that elevation and measures related to the structure itself and the ultimate reservoir surface area can be derived from this document.
15	Project cash flow analysis	Spreadsheet cash flow analysis and forecast for all active PO projects including assumptions on power purchase feed in tariff prices and loan interest rates.
16	Measurement and measuring tools inspection regulation	Dated 24.07.1994
17	Regulation for Electricity Meters	
18	Republic of Turkey Undersecretariat of Treasury. Public Debt Management Report	Dated October 2010.
19	Supporting spreadsheet for calculation of combined margin emission factor for grid-based electricity	Dated 26.08.2011
20	Supporting spreadsheet for investment analysis	Dated 26.08.2011
21	TEIAS database on power production, fuel consumption, commissioning of new plants	2007-2009
22	Turkish GHG Inventory	http://unfccc.int/files/national_reports/annex_i_ghg_inventories/national_inventories_submissions/application/zip/tur-2011-crf-13apr.zip
23	Newspaper article, EPDK license costs	Published on May 2010 with indications of specific prices to acquire project licenses for HEPP
24	Loan agreement Bank B Bank	Loan agreement for the projects Basak, Tuna, Yaprak 2 and Umut 3 dated 24th of August 2009, by Bank B-Bank

25	BANK A termination letter	Letter of BANK A made of May 4th 2009 referencing the initially issued mandate agreement and Letter of Interest both dated April 08, 2008 terminating the loan agreement opportunity by stating that BANK A is no longer intending to enter into financing of the project.
26	BANK A loan refund	SWIFT wire transfer receipt of November 4th 2009 proving that the funds of xxx.xxx.xxx € have been transferred back to BANK A
27	BANK A project finance facility agreement	The Project Finance Facility Agreement between BANK A and PO in the third Draft version of December 21 2008 that was never signed.

The following table summarizes how the documents were used to validate key issues.

Validation criteria	Document references
Project Description	1, 2, 3, 4, 5, 6, 7, 8, 14
Baseline emissions	1, 19, 21, 22
Project start date and crediting period	1, 2, 3, 10
Project ownership	2, 3, 5, 7, 9
Additionality - Investment barrier	1, 2, 3, 4, 10, 11, 12, 13, 15, 18, 20,
Environmental	5, 6, 8, 9
Monitoring	1, 16, 17

2.3 Interviews

The validation team interviewed the following persons:

Yagmur Karabulut / Lale Capalov, representing the PDD developer Mavi Consultants

Ünal Mesci, Nisan Elektromekanik Enerji (hereafter Nisan), Project Manager for all of Nisan's HEPPs

Özden Aydın, Nisan, Assistant to Project Manager

Sertac Toydemir, Civil Engineer of Atac construction company

Ibrahim Altintas Geologist of Atac construction company

Ali Güven, foreman of Atac construction company

2.4 Site Inspections

The site visit took place from 24-28 Mai and included visits to the company headquarters as well as the project site.

2.5 Resolution of Any Material Discrepancy

The validation protocol has been added as Annex to this report. The validation protocol was prepared following the site visit during which all issues identified during the desk review were discussed. All CAR/CLs have been resolved by PPs via provision of additional supporting evidence and appropriate changes to the PDD.

3 VALIDATION FINDINGS

3.1 Project Design

The project is a run-of-river hydropower project located on the Karakus River, Ordu Province, Turkey. The project consists of multiple diversion weirs and a power station with 3 Francis-type turbines with a gross head of 152 m and a project flow of 18 m³/sec. The weirs have a reservoir with roughly 0.114 km² surface area. The turbines have a capacity of 8.15 MWe each, totalling to 24.45 MW, according to the turbine purchase contract. The estimated net power supply to the grid is 68.74 GWh per year, according to the feasibility study. The generated power will be supplied to the national power grid via a 154 kV transmission line. The correctness of the project description was confirmed during the site visit. The description is in line with the feasibility study, the license, the turbine purchase contract, the PIF and the technical drawings reviewed on site.

During the site visit it was determined that the investment decision was taken on 05/06/2008, the date when the turbine contract was signed. According to the contract the turbine capacity was 24.45 MW. According to the feasibility study the expected annual generation is 68.74 GWh. The latter value is in line with the license and a study by Fichtner lead to very similar results.

As per definition of the Turkish EIA regulations the project is a Run-Off-River type HEPP with a small reservoir. The surface area at target water level will be approximately 114,000 m², which results in a power density of more than 200 W/m².

The project location is clearly described in the PDD and in line with the information provided in the feasibility study and the PIF.

Various laws and regulations are applicable prior and during implementation of hydropower projects in Turkey. Table No. 2 of the VCS-PD lists the applicable laws and regulations. The most important documentation is the license issued by the Energy Market Regulatory Authority (EPDK). The license for the project and the EIA-Not Required Statement were issued in 2008. The water utilization agreement was obtained in 2008 and amended in 2010.

According to section 1.1 of the VCS PD the project is owned by Nisan Elektromekanik Enerji San. ve Tic. A.S.. The ownership of the project can be verified by the EPDK license, the DSI water utilization agreement, the turbine purchase contract, the EIA Not Required Statement and the Forest Permit.

3.2 Application of Methodology

3.2.1 Title and Reference

According to the VCS PD, PPs have applied the approved CDM methodology ACM0002 “Consolidated baseline methodology for grid-connected electricity generation from renewable sources” v.12.1. Version 12.1 of the methodology currently in effect. The methodology requires application of the “Tool to calculate the emission factor for an electricity system”, v.02.1.0. and the CDM Additionality Tool, v.5.2.

3.2.2 Applicability

ACM002 is an approved CDM methodology and is explicitly supported by VCS Standard v.3. The project meets the relevant applicability conditions of ACM0002 and the “Tool to calculate the emission factor for an electricity system”.

- The project is a newly installed grid-connected renewable electricity generation project.
- No other power project was operated previously at the site of the project activity;
- The project is a run-of-the-river power plant, and its power density is much higher than 10 W/m²
- The project does not involve co-generation

3.2.3 Project Boundary

The VCS PD defines the project boundary as the Project power plant and all power plants connected physically to the electricity system that the Project power plant is connected to. This is in line with ACM0002.

The relevant baseline emission source is CO₂ emissions from existing and new grid-connected power plants that are replaced by the project activity.

According to the methodology ACM0002 project emissions for hydropower projects have to be considered. They are zero when the power density (Installed capacity / Reservoir surface area) exceeds 10 W/m². According to the VCS-PD the power density exceeds 200 W/m² so that project emissions can be excluded.

3.2.4 Baseline Scenario

ACM0002 defines the baseline as “Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the “Tool to

calculate the emission factor for an electricity system". Baseline emissions are calculated as the product of electricity produced by the project multiplied by the emission factor for the electricity system.

3.2.5 Additionality

The demonstration of additionality follows the CDM Additionality Tool, version 5.2. A three-step process is used, closely matching the requirements of the VCS Standard V.3. where Regulatory Surplus, Implementation Barriers and Common Practice criteria need to be fulfilled.

Step 1: Regulatory Surplus: The construction of hydro-electric power plants at the project location is not mandated by any Turkish law.

Step 2: Implementation Barrier: Investment Analysis

The PPs use an investment analysis to demonstrate the project is not economically attractive in the absence of the VER revenue. The investment analysis is prepared following the Guidance for the assessment of investment analysis".

The validation team reviewed the investment analysis provided as a supporting spreadsheet by the PPs.

Benchmark analysis is applied. This is appropriate since the company has the choice to either invest or not invest. There is no alternative power plant investment possible at the project site. The baseline alternative is power supply from the grid. The project could be implemented by an entity other than PPs. PPs have provided publically available data sources to derive benchmark rate. The required return on equity is based on a World Bank study and the stipulated terms for a medium-term loan are conservative based on the experience from other Turkish renewable energy projects. The WACC approach was correctly applied to identify the IRR benchmark which is in line with the proposed method as per guidelines EB 62. The assumed equity share of 25% is conservative compared with the experience from other Turkish renewable energy projects. In addition it is acknowledged by the Validation Team that actual equity-debt ratio is much higher as was confirmed by the project owner and supported with appropriate evidence [24-27].

Project IRR has been chosen as financial indicator. It is calculated "pre tax" and "at constant prices (no inflation)". This chosen indicator is relevant for the assessment of the project.

The period of assessment is 21 years, including 20 years of operation, the maximum period suggested in the Guidance.

Residual value has been included in the final year of the assessment. The residual value has been calculated based on a 40-year depreciation period for the construction as well as the grid connection and a 20-year depreciation period for the electro-mechanical equipment. This is appropriate for hydro-electric power plants.

Timing of the investment decision is 05.06.2008, i.e. the date when the turbine purchase contract was signed. The validation team has concluded that the key assumptions are traceable and conservatively chosen. In particular,

- power generation is based on the feasibility study, available at that time. The values in the license and the Fichtner study (second lenders engineer study) are very similar.
- power prices are based on the maximum guaranteed price according to the Renewable Energy Law.

- CAPEX are based on the study by the international engineering company Fichtner.
- Operating Expenses (2.5% of capital expenses) are based on the lower bound from a study by the international engineering company Fichtner.
- The development costs (license EPDK, DSI) are based on reasonable assumptions derived from a newspaper article indicating specific costs of 250.000 € per MW, whereas a conservative amount of 2.5 M€ was allocated.
- The sensitivity analysis has been applied with a range of $\pm 15\%$ for all parameters as suggested in EB 62, whereas the benchmark is passed at variations greater than 10% for the power price and the capital expenditure (no pass at variation of O&M)

The probability for the scenarios to occur are considered low as the capital expenditures are evidently higher than indicated (as could be verified ex-post) and power prices remuneration increasing by more than 10% is deemed unlikely.

Step 3: Common Practice Analysis

Run-of-the-river hydro-electric power plants by Independent Power Producers cannot be considered a Common Practice as defined in Chapter 7 of the GHG Protocol for Project Accounting. According to official TEIAS statistics such power plants contribute less than 1% of the total power production in Turkey. They can therefore not be considered as a predominant technology in the Turkish power sector.

3.2.6 Quantification of GHG Emission Reductions and Removals

In section 3 of the VCS-PD emission reductions are calculated in accordance with equation (1) of AMS-I.D:

$$BE_y = EGBL_y * EFCO2_{grid,y}$$

Where power generation (EG) has been taken from the feasibility study and the emission factor has been calculated correctly according to the Tool to calculate the emission factor for an electricity system. In particular PPs have correctly applied the following steps:

Step 1: The relevant electricity system was correctly identified as Turkey's national grid. There are no price differences and no transmission constraints in the project area.

Step 2: PP have correctly chosen Option 1, i.e. to exclude off-grid power plants. This is appropriate for the case of Turkey.

Step 3: PP have correctly chosen the Simple OM method. They have demonstrated that the share of low-cost resources is below 50%.

PPs have correctly chosen the ex-ante option to calculate the Simple OM. The most recent data is 2007 to 2009.

Step 4: PP have correctly chosen Option B. The operating margin emission factor is calculated based on power sector emissions in the national GHG inventory divided by the total power production based on TEIAS data. The emissions in the GHG inventory have been calculated based on the total fuel consumption in the electricity system, which have been multiplied with national default values for net calorific values and fuel-specific emission factors for liquid, solid and gaseous fuels with emission factors

of 73, 98, 56 t/TJ. Using national default factors is the preferred option on pages 20, 21 of the Tool. The validation team considers the calculation method as in line with option B in step 4 of the Tool. For the preparation of the GHG inventory the total fuel consumption and total electricity generation of the electricity system was available. The validation team has confirmed that using this calculation method does not result in systematically higher values for the operating margin emission factor and thus does not negatively impact the conservativeness of the calculations.

Step 5: PP have correctly chosen the group of power plants to be included in the BM. There are more than 5 power plants in the 20% of power generation volume that have been built most recently. PPs have chosen Option 1 (ex-ante option). Only power plants commissioned between 2004 and 2009 have been included.

Step 6: Equation (13) from the Tool is correctly applied.

Efficiency factors have been chosen in a transparent and conservative manner:

Fuel Oil / Naphta / Diesel Oil: Default efficiency factor for combined-cycle plants has been applied. This is conservative.

Natural gas: Default efficiency factor for combined-cycle plants has been applied. This is conservative.

Coal & Lignite: Efficiency factors of 40% have been applied. PPs have demonstrated in a separate spreadsheet that this conservative.

Emission factors at lower end of range were chosen as required by the Tool.

Fuel types for newly constructed plants have been assigned conservatively. The more conservative option was chosen when different fuels are used at a given power plant.

Step 7: Equation (14) from the Tool is correctly applied.

Weights have been correctly chosen as $w_{OM}=0.5$ and $w_{BM}=0.5$.

The Validation Team therefore was able to verify the EF for the OM as 0.653 and the BM to be 0.450. As a result the CM is verified to be 0.551.

For renewable energy projects as per methodology ACM0002 the leakage emissions are considered zero because the emissions sources can be neglected.

Applying the emission factors and the estimated annual net electricity generation as explained in section 3.1 above Emission Reductions of $(32,600 \text{ MWh} * (0.653 * 0.5 + 0.450 * 0.5) \text{ t / MWh}) = 17,960 \text{ tCO}_2$ are expected to occur.

3.2.7 Methodology Deviations

None

3.2.8 Monitoring Plan

According to the VCS-PD ACM0002 is used as monitoring methodology. This is in line with VCS v.3. The methodology requires the following parameters that are applicable to be included in the project activity

EG_{facility,y}: Quantity of net electricity generation supplied by the project plant/unit to the grid in year y.

EF_{grid,CM,y}: Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system”.

According to section 4 of the VCS-PD EG_{facility} will be recorded by two redundant metering devices continuously. The meters will be read each month by TEIAS employees. A metering protocol will be signed by both TEIAS and a plant employee. These records will provide the data for the monthly invoicing to TEIAS and are accessible via the official PMUM website. The data from the PMUM website will be used for the monitoring report. The proposed measurement method is appropriate and in line with standard industry practice in Turkey.

Metering devices need to meet the technical requirements of the Regulations for electricity meters. Metering devices are technically checked and certified by independent companies for their accuracy. After installation, TEIAS tests both metering devices in order to ensure that the devices satisfy all requirements and are calibrated properly.

Maintenance and calibration of the metering devices will be made by TEIAS periodically. If there is a noticeable difference between the readings of two devices, maintenance and tests of the metering devices and the associated equipment will be done before waiting for the periodical maintenance.

3.3 Environmental Impact

The project activity is covered by a preliminary environmental impact analysis (PIF). A full EIA is not required since the scale of the project is below the applicable. The PIF has not identified any significant environmental impacts. The validation team has confirmed during the site visit that the weir will be equipped with a fish ladder and that a minimum flow will be released to the river bed.

3.4 Comments by stakeholders

Stakeholder consultations for run-of-the-river hydro-electric power plants are not required by Turkish law. Nevertheless the project company has contacted and informed the local government during the planning phase. Local residents are impacted by the project in two major ways. First, some private land is expropriated or bought by the project company and local residents are very interested in the expropriation process and the compensation amounts. Second, some local residents are able to find employment during the construction and operation phases of the project. The water in the project area is not used for any irrigation activities so that resource conflicts are limited.

4 VALIDATION CONCLUSION

Nisan Elektromekanik Enerji San. ve Tic. A.S.

Cinnah Caddesi. Ahenk Sokak.

No:5 / 2 Çankaya in Nakara,

Turkey

23 September, 2011

RE: Validation Statement – Umut-II Hydropower Plant, Turkey under VCS, first crediting period from October 1st, 2012

Nisan Elektromekanik Enerji San. ve Tic. A.S. with registered office at Cinnah Caddesi. Ahenk Sokak. No:5 / 2 Çankaya in Nakara, Turkey has engaged TÜV Rheinland Energie und Umwelt GmbH to review and validate the VCS PD and all assertions related to the GHG project under VCS v.3. The GHG assertions, indicated in the VCS PD issue 23.09.2011 v.1.4 indicate annual achievable emission reductions of 37,883 t CO₂e per annum.

We have conducted the validation of the GHG assertion in accordance with the requirements of the VCS program requirements, the standard ISO 14064-3 and approved UNFCCC methodology ACM0002, v.12.1.0. to a reasonable level of assurance by applying a materiality threshold of 5 %. The project information validated includes the VCS PD delivered to TÜV Rheinland Energie und Umwelt GmbH and all relevant information and evidence acquired during the validation process as stated in the validation report made of 23. September 2011.

We have come to the conclusion that based on our review and all available documentation the GHG assertion is made in accordance with the requirements of the VCS program and is material correct and fairly represents the GHG emissions data and information without material discrepancies.

Cologne, 23 September 2011



Roland Wollenweber

5 APPENDIX A

Appendix A

VCS VALIDATION PROTOCOL

- FOR VCS STANDARD V.3 PROJECTS -

Project Title and Location:

Umut-II Hydropower Plant, Turkey

REPORT NO. 2121 6157c

VERSION NO. 01

Table 1: Validation requirements

(Based on VCS Standard, V.3 and § 37 of the CDM Modalities and Procedures and on CDM Validation and Verification Manual, Annex 3 of EB44)

Checklist question	Ref.	MoV ¹	Findings, comments, references, data sources	Draft conclusion	Final conclusion
1. VCS Project Description Document					
1.1. Is the VCS PD presented for validation based on the latest template available at the VCS website? <i>Indicate Yes / No answer and describe all inconsistencies in the Tables 2 and 3 accordingly.</i>		DR	Yes	OK	OK

¹ MoV = Means of Verification, DR = Document Review, I = Interview, www = internet search.

1.2. Has the VCS PD been established in accordance with the VCS requirements for completing VCS PD issued by the VCS Association?		DR	Yes, see 1.1.	OK	OK
2. Project Description					
2.1. Does the VCS PD contain a description, which provides the reader with a clear understanding of the precise nature of the project activity and the technical aspects of its implementation?	1.4, 1.9	DR, site visit	Yes, the project description in section 1.1 and 1.8 of the VCS PD is clear and comprehensive. The project is a run-of-river hydropower project located on the Karakus River, Ordu Province, Turkey. The project consists of multiple diversion weirs and a power station with 3 Francis-type turbines with a gross head of 152 m and a project flow of 18 m³/sec. The weirs have a reservoir with roughly 0.114 km² surface area. The turbines have a capacity of 8.15 MWe each, totalling to 24.45 MW, according to the turbine purchase contract. The estimated net power supply to the grid is 68.74 GWh per year. The generated power will be supplied to the national power grid via a 154 kV transmission line. The correctness of the project description was confirmed during the site visit.	CL01 CL02 CL03 CAR01	OK

2.2. In the case of green-field project activity, is the project design described sufficiently by means of specifications, drawings and manuals? <i>Provide Yes/No answer and indicate the documents which have been reviewed in relation to the issue.</i>	1.4, 1.9	DR, site visit	Yes. Section 1.1 and 1.8 contain specifications for the works and equipment as well as maps of the weir, tunnels and powerhouse in order to describe the project activity. The validation team was able to confirm the accuracy of the project description by reviewing maps and technical drawing during the site visit as well as via the on-site inspection.	OK	OK
---	---------------------	-------------------------------	---	-----------	-----------

2.3. Does the project activity reflect current good practices, uses state of the art technology or would the technology result in a significantly better performance, than any commonly used technologies in the host country? <i>Provide the description of how validation has been carried out and what comparisons have been made.</i>	1.4, 1.9	DR	Yes. The project uses Francis type turbines, which are commonly used for hydro-electric power production. Francis-type turbines are appropriate for hydro-electric power plants with a wide range of design discharges at the total head of 78 m. The validation reviewed the feasibility study as well as maps and drawings during the site visit.	OK	OK
---	-------------	----	---	----	----

2.4. In cases where the project activity involves the alteration of an existing installation or process, does the VCS PD provide a clear description of the differences between the project and the pre-project scenario? <i>Please, provide Yes/Now answer and update Tables 2, 3 and 4 accordingly, if there is anything unclear in the provided description.</i>		DR	Not applicable. The project is a greenfield project activity.	OK	OK
2.5. Does the project grouping apply for the proposed project? If yes, does the VCS PD include a description of one central GHG information system and controls associated with the project and monitoring?		DR	Not applicable. No project grouping is planned for the project activity.	OK	OK

2.6. Does the central GHG information system and controls include items identified in ISO14064-3:2006, clause 4.5?		DR	Not applicable. No project grouping is planned for the project activity.	OK	OK
2.7. How has the sampling been determined? Have any of the sub groups and associated activities within each sub group been considered? Does the sampling meet the requirement set out under IAF Guide to ISO/IEC 66/1999 at a minimum?		DR	Not applicable. No project grouping is planned for the project activity. The only monitoring parameter EG_{facility} is monitored continuously. No sampling is applied.	OK	OK
3. Project Start Date and Crediting Period					

3.1. Are the project's start date clearly defined and evidenced? Does the project start date meet the requirements as stated in VCS 2007.1 clause 5.2.1? If project grouping is applied, each grouped project shall meet the requirement of VCS 2007.1 clause 5.2.1.	1.6, 7	DR	In section 1.5 of the VCS PD the project start date is identified as 01/01/2013, i.e. the expected commissioning date. The definition of the project start date according to the VCS Standard (page 7) is correctly applied. The project start date is after January 2002 and therefore meets the requirements of clause 5.2.1. No project grouping is applied.	OK	OK
3.2. Is the start of the crediting period clearly defined and reasonable?	1.6, 7	DR	Yes, according to sections 1.6 of the VCS PD the start date of the crediting period is 01/12/2013. This coincides with the project start date and is reasonable.	OK	OK
4. Baseline and Monitoring methodology					
4.1. General requirements					

4.1.1 Is the methodology used in the project activity an approved VCS Program methodology or a methodology from a VCS-approved GHG program? Is the selected methodology version still valid?	2.1	DR	The applied methodology ACM0002 is a consolidated CDM methodology, which counts as a VCS approved GHG program. Version 12.1 of the methodology ACM0002 is still valid.	OK	OK
4.2. Applicability of the selected methodology					
4.2.1 Are all applicability conditions of the selected baseline and monitoring methodology and all tools involved satisfied by the project activity? <i>Please indicate Yes/No answer. In case of positive conclusion provide details of the validation measures. Otherwise amend the Tables 2 and 3</i>	2.2	DR	Yes. The project meets the relevant applicability conditions of ACM0002 and the “Tool to calculate the emission factor for an electricity system”. • The project is a newly installed grid connected renewable electricity generation project. • No other power project was operated previously at the site of the project activity; • The project is a run-of-the-river power plant, and its power density is much higher than 10 W/m² • The project does not involve co-generation	OK	OK

<i>accordingly.</i>					
4.2.2 Is the selection of the applied baseline and monitoring methodology justified?	2.2	DR	Yes, the project fulfills all applicability conditions. There is no other approved CDM or VCS methodology that is applicable to the project. See 4.2.1.	OK	OK
4.2.3 Is the selected methodology correctly quoted in all related documents?		DR	Baseline definition is quoted correctly on pages 13 and 15 of the VCS-PD.	OK	OK

4.2.4 Does the VCS PD sufficiently describe all the GHG emission sources or sinks occurring as a result of project activity, which have not been accounted for under the selected methodology and are expected to contribute more than 1% of the overall expected average annual emission reductions? <i>Provide Yes/No answer. Indicate the sources or sinks of GHG, which were proved to be negligible. Otherwise amend the Tables 2 and 3 accordingly.</i>	2.4	DR	Yes. The power density of the project is 215 W/m². This is substantially higher than the threshold of 10 W/m². Methane emissions from the project can therefore be considered as negligible in accordance with the applied methodology. No other project emissions, accounting for more than 1 % of the annual emission reductions are expected.	OK	OK
4.3 Project boundary					

4.3.1 Does the VCS PD correctly describe the project boundary? <i>Provide Yes/No answer. And amend the Tables 2 and 3 if needed.</i>	3.2	DR	According to ACM0002 “the spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the CDM project power plant is connected to.”	CL04	OK
4.3.2 Does the VCS PD correctly indicate and describe the emission sources and sinks of GHG gases that are included in the project boundary?	2.3	DR	In section 2.3 of the VCS PD the sources and gases included in the project boundary are identified. The list corresponds to Table 1 of ACM0002. Project emissions of methane from the reservoir have been considered. The project emissions have been set equal to 0, in accordance with the methodology (page 8), since the power density far exceeds the threshold of 10 W/m².	CAR02 CL05	OK

4.3.3 In cases where the methodology allows project participants to choose whether a source or gas is to be included in the project boundary, is the choice explained and justified by PPs?		DR	Not applicable. ACM0002 does not offer such a choice.	OK	OK
4.4 Baseline identification					
4.4.1 Has the procedure contained in the selected methodology to identify the most reasonable baseline scenario been applied correctly and documented in the VCS PD?		DR	Not applicable. ACM0002 (page 4) clearly defines the baseline as: "Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the Tool to calculate the emission factor for an electricity system. The methodology offers no further procedure to identify the most reasonable baseline scenario.	OK	OK

4.4.1.1 Is the identified baseline scenario plausible?		DR	Yes. The selected baseline scenario is plausible for the following reasons. The baseline scenario complies with existing legislation and policy. The PPs are under no obligation to implement the project. The project site is not a plausible location for an alternative power plant. As shown in section 2.5 the project is not attractive on financial grounds without the VCU revenue.	OK	OK
4.4.1.2 Are all assumptions stated in a transparent and conservative manner?	4.2	DR	Yes. Fuel types and efficiency factors for existing power plants and Build Margin plants have been chosen in a conservative manner. See 4.5.1.	OK	OK
4.4.2 Does the selected methodology require the use of tools <u>and</u> does VCS PD reflect that correctly?	2.5, 4.2	DR	Yes. ACM0002 requires the use of two tools. (1) The “Tool to calculate the emission factor for an electricity system” and (2) the “Tool for the demonstration and assessment of additionality”; The relevant tools have been applied in sections 2.5 and 3.1.	OK	OK
4.4.2.1 Were all the tools applied correctly?	2.5, 4.2	DR	The “Tool to calculate the emission factor for an electricity system” has been correctly applied in a step-by-step fashion in section 3.1 of the VCS-PD. See findings to 4.5.1. The Additionality Tool has been correctly applied in a step-by-step fashion in section 2.5 of the VCS-PD.		

4.4.3 In case the methodology requires several alternative scenarios to be considered in the identification of the most reasonable baseline scenario, have all scenarios been considered <u>and</u> have no reasonable alternative scenario been excluded?		DR	Not applicable. ACM0002 does not require the consideration of alternative baseline scenarios for this project type.	OK	OK
4.4.3.1 Has the choice of the baseline scenario been done using conservative assumptions?		DR	Not applicable. ACM0002 does not require the consideration of alternative baseline scenarios for this project type.	OK	OK

4.4.4 Is the identified baseline scenario reasonable according to the assumptions, calculations and rationales used in the VCS PD and other reference sources?		DR	Not applicable. ACM0002 (page 4) clearly defines the baseline and does not require the consideration of alternative baseline scenarios for this project type.	OK	OK
4.4.5 Does the VCS PD describe how the national and sectoral policies relevant to the baseline scenario have been identified and considered in the VCS PD?	2.5	DR	Table 10 in section 2.5 of the VCS PD identifies the laws and regulations that apply to the Electricity Generation sector. The following laws are included: 1. Electricity Market Law 2. Law on Utilization of Renewable Energy Resources for the Purpose of Generating Electricity Energy 3. Energy Efficiency Law 4. Forest Law 5. Environment Law Both the baseline and the project are consistent with the above laws.	OK	OK

4.4.6 Does the VCS PD provide a verifiable description of the identified baseline scenario, including a description of the technology that would be employed and/or the activities that would take place in the absence of the project activity?	2.4	DR	In section 2.4 the baseline scenario is identified in accordance with ACM0002 (page 4): "Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the Tool to calculate the emission factor for an electricity system." The methodology offers no further choice regarding the baseline scenario.	OK	OK
4.5 Algorithm and/or formulae used to determine emission reductions					

4.5.1 Are all calculations applied and documented according to the selected methodology and in a complete and transparent manner?	4.2	DR	In section 3.1 of the VCS PD version 2 of the “Tool to calculate the emission factor for an electricity system” is applied step by step. Step 1: The relevant electricity system was correctly identified as Turkey’s national grid. There are no price differences and no transmission constraints in the project area. Step 2: PP have correctly chosen Option 1, i.e. to exclude off-grid power plants. This is appropriate for the case of Turkey. Step 3: PP have correctly chosen the Simple OM method. They have demonstrated that the share of low-cost resources is below 50%. The relevant data is in Table 15. PP have correctly chosen the ex-ante option to calculate the Simple OM. The most recent data is 2006 to 2008. Step 4: PP have correctly chosen Option B, since there are no power plant-specific emission factors available. Equation (7) from the Tool is correctly applied. Step 5: PP have correctly chosen the group of power plants to be included in the BM. There are more than 5 power plants in the 20% of power generation volume that have been built most recently. PP have chosen Option 1 (ex-ante option). Only power plants commissioned between 2004 and 2008 have been included. Step 6: Equation (13) from the Tool is correctly applied. Efficiency factors have been chosen in a transparent and conservative manner: Fuel Oil / Naphta / Diesel Oil: Default factor for combined-cycle plants: 46%. Natural gas: Default factor from Tool for combined-cycle plants:	CAR03, CL06	OK
v3.0			32		

4.5.1 (continued)	4.2	DR	Emission factors at lower end of range were chosen as required by the Tool. Fuel types for newly constructed plants have been assigned conservatively. The more conservative option was chosen when different fuels are used at a given power plant. Step 7: Equation (14) from the Tool is correctly applied. Weights have been correctly chosen as $w_{OM}=0.5$ and $w_{BM}=0.5$.	CAR03 CL06	OK
4.5.2 In case the methodology allows a selection between different options for equations or parameters, has adequate justification been given and have the correct equations and parameters been used, in accordance with the methodology selected?	4.2	DR	Yes. See 4.5.1.	OK	OK

4.5.3 In case some data and parameters will not be monitored throughout the crediting period, but have already been determined and fixed, are all data sources, assumptions and calculations correct, applicable to the proposed VCS project activity and conservative?	4.2	DR	The one factor not monitored throughout the crediting period is the Combined Margin emission factor. The factor has been calculated correctly, and the assumptions are conservative, see also 4.5.1	OK	OK
4.5.4 In case data and parameters will be monitored on implementation and hence become available only after validation of the project activity, are the estimates provided in the VCS PD for these data and parameters reasonable?		DR, I	The key monitoring parameter is net electricity generation, which is estimated at 68,740 MWh per year. In the VCS PD it is stated that the source of the data is the feasibility study prepared by Coyne & Bellier.	OK	OK

4.5.5 Have the major risks and uncertainties, which can influence the emission reduction estimates, been identified and addressed in the VCS PD?	1.11	DR	Section 3.4 of the VCS-PD identifies the water flow and precipitation levels as the major risk to the emission reduction estimates. This is appropriate for a hydro power project.	OK	OK
4.6 Leakage					
4.6.1 Has the leakage been identified and calculated according to the approved methodology?	4.3	DR	Not applicable. No leakage emissions are considered in ACM0002. Moreover, the energy generating equipment is not transferred from another project site. The validation team has reviewed the turbine purchase contract.	OK	OK
4.6.2 Have the leakage been addressed in complete, conservative and substantiated manner?		DR	Not applicable. See 4.6.1.	OK	OK
4.6.3 Are uncertainties in the leakage emission estimates properly addressed?		DR	Not applicable. See 4.6.1.	OK	OK
5. Additionality					

5.1 Identification of alternatives					
5.1.1 What additionality tool has been selected in the VCS PD? Does it comply with the requirement of VCS 2007.1 clause 5.8 and the latest FAQs at VCS website? (http://www.v-c-s.org/faq.html)	2.5	DR	It is stated in section 2.5 of the VCS PD, that the CDM Additionality Tool, version 5.2 has been applied. This is in compliance with ACM0002. A three-step process is applied, which meets the requirements of the project test in clause 5.8 of the VCS 2007.1. (1) Identification of Alternatives: It is shown that the baseline is also consistent with current laws. (Regulatory Surplus) (2) Investment Analysis (Implementation Barriers) (3) Common Practice Analysis (Common Practice)	OK	OK
5.1.2 Does the VCS PD identify and list credible alternatives to the VCS project activity in order to determine the most realistic baseline scenario, unless selected approved methodology prescribes/identifies the baseline scenario and no further analysis is required?	2.5	DR	Three alternatives have been listed under section 2.5. Scenario 3 has been excluded. The project site is not appropriate for any other power plant project.	CAR04	OK

5.1.3 Does the list of alternatives include as one of the options that the project activity is undertaken without being registered as a VCS project activity?	2.5	DR	Yes. This is alternative 1.	OK	OK
---	-----	----	-----------------------------	----	----

5.1.4 Does the list contain all realistic/credible alternatives that the DOE, on the basis of its local and sectoral knowledge, considers to be viable means of supplying the outputs or services that are to be supplied by the project activity? <i>Note: All alternatives listed in the selected methodology should be included, as well as those not covered by the methodology.</i>	2.5	DR	There are no alternative ways to generate power at the project site. The baseline proscribed by the methodology, i.e. the increased supply of grid-based electric power by existing and new sources covers all alternatives that would provide the same output as the project.	OK	OK
--	-----	----	---	----	----

5.1.5 Is the exclusion of the alternatives for legal reasons justified? <i>Note: Some alternatives might be illegal, according to the local regulations, but still widely practiced due to lack of enforcement. It should be verified.</i>	2.5	DR	Not applicable. No alternatives have been excluded for legal reasons.	OK	OK
5.2 Investment Analysis					
5.2.1 Are all sources of revenues (including savings) have been considered in the VCS PD and all calculations?	2.5	DR	Yes. The investment analysis considers electricity sales revenue, which is the only relevant revenue of the project.	OK	OK
5.2.2 Is the type of investment analysis selected correctly in the VCS PD?	2.5	DR	Yes. Benchmark analysis is applied. This is appropriate since the company has the choice to either invest or not invest. There is no alternative power plant investment possible at the project site.	OK	OK

5.2.3 Is the selected financial indicator chosen and applied correctly?	2.5	DR	Yes. It is stated in section 2.5 (sub-step 2b) that project IRR has been chosen as financial indicator. It is calculated “pre-tax” and “at constant prices (no inflation)”	OK	OK
5.2.4 Is the guidance on IRR calculation and assessment correctly applied? <i>Note: Means of validation should be recorded.</i>	B.5	DR	(1) Project IRR has been chosen as financial indicator. It is calculated “pre tax” and “at constant prices (no inflation)”. This chosen indicator is relevant for the assessment of the project. (2) The period of assessment is 21 years, including 20 years of operation, the maximum period suggested in the Guidance. (3) Residual value has been included in the final year of the assessment. The residual value has been calculated based on a 40-year depreciation period for the construction and a 20-year depreciation period for the electro-mechanical equipment. This is appropriate for hydro-electric power plants. (4) The timing of the investment decision is 05.06.2008, i.e. the date when the turbine purchase contract was signed. The validation team has concluded that the key assumptions are traceable and conservatively chosen. In particular, • power generation is based on the feasibility study. The values in the license and the Fichtner study are very similar. • power prices are based on the maximum guaranteed price according to the Renewable Energy Law. • CAPEX are based on the study by the international engineering company Fichtner. • Operating Expenses (2.5% of capital expenses) are based on the lower bound from a study by the international engineering company Fichtner.	CAR05 CAR06 CAR07 CAR08 CAR09 CAR10 CL07	

5.2.4 (continued)	2.5	DR	(5) PP have provided IRR calculation spreadsheet (6) Loan interest and pay back are not considered as outflows in the IRR calculation. (7) Project could be implemented by entity other than PPs. The required return on equity is based on a World Bank study and the stipulated terms for a medium-term loan are conservative based on the experience from other Turkish renewable energy projects. The assumed equity share of 25% is conservative compared with the experience from other Turkish renewable energy projects. (8) The baseline alternative is power supply from the grid, therefore benchmark analysis is appropriate. (9) Weighted cost of capital has been calculated, which is an appropriate benchmark for project IRR.		
5.2.5 In case project participants use values from Feasibility Study Reports (FSR) is it possible to verify that the period between the FSR date and investment decision was reasonably short and FSR values did not change materially?	2.5	DR	CAPEX/OPEX are based on the Fichtner study. Power production is based on the feasibility study by Coyne & Bellier. The time between the feasibility studies and the investment decision was less than one year.	OK	OK

5.2.6 Are all the values consistent between FSR and VCS PD <u>and</u> are inconsistencies properly justified?			Yes	OK	OK
5.2.7 Were all the values from FSR applicable and valid at the time of the investment decision?			Yes	OK	OK
5.2.8 Is it reasonable to assume that no investment would be made at a rate of return lower than the benchmark by, for example, assessing previous investment decisions by the project participants or some verifiable circumstances that have lead to a change in the benchmark?	2.5	DR	Project could be implemented by entity other than PPs, so that publically available benchmarks have to be chosen. The required return on equity is based on a World Bank study and the stipulated terms for a medium-term loan are conservative based on the experience from other Turkish renewable energy projects. The assumed equity share of 25% is conservative compared with the experience from other Turkish renewable energy projects.	OK	OK

5.2.9 Is the Investment Analysis prepared in compliance with the latest version of the “Guidance on the Assessment of Investment Analysis” as provided by the CDM EB?			See findings to 5.2.4		
5.3 Barrier analysis					
5.3.1 Are there any issues addressed in the barrier analysis that have a clear impact on the financial viability of the project activity and that shall be assessed by an investment analysis?	2.5	DR	Not applicable. Barrier analysis is not applied.	CL08	OK

5.3.2 Do the listed barriers exist <u>and</u> is their existence substantiated? Note: (a) by independent sources of data such as relevant national legislation, surveys of local conditions and national or international statistics and/or (b) by interviews with relevant individuals: including members of industry associations, government officials or local experts if necessary?	2.5	DR	Not applicable	CL09	OK
--	-----	----	----------------	------	----

5.3.3 Would any of the identified barriers prevent the implementation of the project activity but not equally prevent the implementation of the possible alternatives, in particular the implementation of the identified baseline scenario?	2.5	DR	Not applicable	OK	OK
5.4 Common practice analysis					
5.4.1 Does the additionality tool require common practice analysis for the project?	2.5	DR	Yes. The CDM Additionality Tool requires a common practice analysis.	CAR11	OK
5.4.2 If the PPs claim in the VCS PD that VCS project activity is the “first of its kind”, is it justified?	2.5	DR	Not applicable. The PPs do not claim that the project is “first of its kind”.	OK	OK

5.4.3 Are the geographical scope of the project activity identified correctly?	1.5, 2.5	DR	The project location is correctly identified in section 1.9. Only hydropower plants in the same DSI region are considered as similar project activities.	OK	OK
5.4.4 Does the VCS PD provide an explanation why this region was selected and deemed more appropriate <u>and</u> is this explanation traceable and reliable?		DR	DSI categorizes Turkey in certain hydrological regions based on their geographical, precipitation, geological and managerial criteria.	OK	OK

5.4.5 Are there similar operational project activities, other than VCS activities, “widely observed and commonly carried out” in the defined region? <i>Note: Use official sources and local and industry expertise.</i>		DR	According to TEIAS statistics, quoted in section 3.1 of the VCS-PD, hydro power plants account for 18.7% of power production in Turkey. However, most of the generation comes from large state-owned hydro-electric projects. Moreover, the share of hydro power in Turkey has been decreasing from a peak of approx. 40%, while the share of thermal generation has been increasing. The share of independently owned hydroelectric power plants is less than 1%, many of the projects are developed as VER projects and therefore the project type cannot be considered to be common practice as defined in Chapter 7 of the GHG Protocol for Project Accounting. (See Section 5.8 of the VCS 2007.1). In the VCS-PD, similar project activities are defined as hydroelectric power plants with the following characteristics: (a) generation license from EMRA, (b) run by IPPs, (c) same DSI region and (d) similar capacity and load factor.	OK	OK
--	--	----	--	----	----

5.4.6 In case there are similar commercially operated project activities, other than VCS activities, already “widely observed and commonly carried out” in the defined region, are there essential distinctions between the VCS project activity and the other similar activities?			In the VCS-PD there is only one similar project activity that is not a VER project. However, there is an important difference in that the Umut-II project has four weirs and correspondingly high construction costs.	OK	OK
6. Monitoring plan					
6.1 Are all parameters required by the selected approved methodology or tool identified <u>and</u> listed in the VCS PD?	3	DR	ACM0002 only requires the monitoring of EG _{facility} , which is identified and listed in section 4. CapPJ and APJ do not have to be monitored if CH ₄ emissions have been demonstrated to be negligible due to the small reservoir size. These emissions should be discussed in table 8 and excluded as negligible.	CL10	OK

6.2	Is the measurement method clearly stated for each value to be monitored and deemed appropriate?	3	DR	According to section 4 of the VCS-PD, EG_{facility} will be recorded by two redundant metering devices continuously. The meters will be read each month electronically by TEIAS employees. A protocol will be printed on-site and signed by both TEIAS and a plant employee. These records will provide the data for the monthly invoicing to TEIAS and are accessible via the official PMUM website. The data from the PMUM website will be used for the monitoring report. The proposed measurement method is appropriate and in line with standard industry practice in Turkey.	OK	OK
6.3	Are values of the ex-ante parameters / monitoring parameters selected correctly and conservative in accordance to methodology or tools?	4.2	DR	The only ex-ante parameter is $EF_{\text{grid,CM,y}}$. The value of the parameter has been correctly derived in section 3 of the VCS-PD. The only monitoring parameter is EG_{facility}. The expected value of the parameter is based on the study by Fichtner.	CAR01	OK
6.4	Is the measurement equipment for each parameter described and deemed appropriate?	3	DR	The metering devices are under the control by TEIAS. They need to meet the technical requirements set out by the relevant legislation, i.e. the Communiqué for Metering Devices to be used in the Electricity Market.	OK	OK

6.5	Is the measurement accuracy addressed and deemed appropriate?	3	DR	The metering devices need to fulfil the technical requirements set out by the Communiqué for Metering Devices to be used in the Electricity Market. The Communiqué describes the minimum accuracy requirement the metering devices have to fulfil, which are categorized according to the installed capacity. As the rated output of the Project Activity is between 10 MW and 100 MW, as this communiqué sets out, each metering device has to have accuracy equal to or better than 0.5%.	OK	OK
6.6	Are procedures in place on how to deal with erroneous measurements <u>and</u> are the corrective actions identified?	3	DR	There are two redundant meters in place to measure EGfacility. In case the main meter fails the back-up meter is used. Erroneous measurements can be dealt with on a monthly basis. Each month, an officer from TEİAŞ and the plant manager/electricity technician of the plant will record the reading and sign the record.	OK	OK
6.7	Is the frequency of measurement identified and deemed appropriate?	3	DR	According to section 3 of the VCS PD EGfacility is measured continuously. Monthly meter readings are taken. This is appropriate and in line with standard industry practice in Turkey.	OK	OK
6.8	Is the monitoring plan documented according to the approved methodology and in a complete and transparent manner?	3	DR	EGfacility is measured according to ACM0002. Meter readings are electronically converted into invoices. The amounts are therefore identical.	OK	OK

6.9 Are the sampling, measurement methods and procedures defined?	3	DR	Yes, see 6.2.	OK	OK
6.10 Are procedures identified for maintenance of monitoring equipment and installations?	3	DR	Maintenance of the metering devices will be made by TEIAS periodically according to the relevant national regulations. If there is a noticeable difference between the readings of two devices, maintenance and tests of the metering devices and the associated equipment will be done before waiting for the periodical maintenance.	CL11	OK
6.11 Are the equipment calibration intervals identified and justified?	3	DR	Metering devices need to meet the technical requirements of the relevant national regulations. Metering devices are technically checked and certified by independent companies for their accuracy. After installation, TEIAS tests both metering devices in order to ensure that the devices satisfy all requirements and are calibrated properly. According to Section 3 of the VCS-PD calibration of the metering devices will be made by TEIAS periodically. If there is a noticeable difference between the readings of two devices, maintenance and tests of the metering devices and the associated equipment will be done before waiting for the periodical calibration and maintenance.	CL12	OK

6.12 Are procedures identified for day-to-day records handling (including what records to keep, storage area of records and how to process performance documentation)?	3	DR	On a monthly basis data will be downloaded from the PMUM website and transferred into the monitoring report spreadsheet for the emission reduction calculation. Scans or hard copies of monthly PMUM reports will also be archived.	OK	OK
6.13 Are the monitoring arrangements described in the monitoring plan feasible within the project design?	3	DR	Yes, the monitoring arrangements are part of the regular project operation.	OK	OK
6.14 Are the means of implementation of the monitoring plan, including the data management and quality assurance and quality control procedures, sufficient to ensure that the emission reductions achieved by / resulting from the project activity can be reported ex post and verified?	3	DR	Yes, see 6.1 to 6.13.	OK	OK

6.15 Do the PPs make provisions for personnel training needs?	3	DR	The monitoring arrangements are part of the regular project operation. No special training is necessary.	OK	OK
6.16 Is the authority and responsibility of overall project management clearly described?	3	DR	The Project Owner has assigned Ms. Özden Aydın as Carbon Manager for the project.	CL13	OK
6.17 Are procedures identified for emergency preparedness for cases where emergencies can cause unintended emissions?			Not applicable. Due to the absence of any combustion processes, unintended emissions are not likely for hydroelectric power plants.	OK	OK
6.18 Are procedures identified for review of reported results/data?	3	DR	According to section 3, each month, an officer from TEİAŞ and the plant manager/electricity technician of the plant will record the reading and sign the record.	OK	OK

6.19 Is the data archiving period for this project activity stated in the VCS PD and appropriate? <i>Note: All archived monitoring data, required for verification and issuance, should be kept for at least two years after the end of the crediting period or the last issuance of CER.</i>	3	DR	According to section 3, all monitoring data will be archived electronically and on paper until at least 2 years after the end of the last crediting period.	OK	OK
7. Monitoring of the leakage					
7.1 Does the monitoring plan provide for the collection and archiving of all relevant data necessary for determining leakage?		DR	Not applicable. According to ACM0002, no leakage is considered.	OK	OK

7.2	Is the choice of project leakage indicators made according to selected methodology in a reasonable and conservative manner? <i>Note: local knowledge and sectoral expertise shall also be considered.</i>			Not applicable. See 7.1.	OK	OK
7.3	Is the measurement method clearly stated and deemed appropriate for each leakage value?			Not applicable. See 7.1.	OK	OK
8. Sustainable development						
8.1	If VCS PD indicates any additional environmental benefits of the project, other than GHG emission reductions, were those benefits properly substantiated?	1.4	DR	No additional environmental benefits are claimed in the VCS-PD	CL14	OK
9. Stakeholders' consultation and comments						

9.1 Were the stakeholders identified in appropriate and complete manner?	6	DR	Comments from local stakeholder were invited via informal consultations. A comprehensive Project Information File was prepared and submitted to the Ministry of Environment and Forestry. The Ministry concluded that there were no significant environmental impacts and that a full EIA was not necessary.	OK	OK
9.2 Are the identified stakeholders plausible?	6	DR	Yes. The impacts of the project are local, so local residents are the key stakeholders. The Ministry is the key stakeholder for the impacts on forest land and environmental impacts.	OK	OK
9.3 Does VCS PD describe the means being used to invite local stakeholder's comments?	6	DR	Comments from local stakeholder were invited via informal consultations. A comprehensive Project Information File was prepared and submitted to the Ministry of Environment and Forestry. The Ministry concluded that there were no significant environmental impacts and that a full EIA was not necessary.	OK	OK
9.4 Were those means appropriate?	6	DR	Yes. The project does not have any significant environmental impacts.	OK	OK
9.5 Was the project presented to the stakeholders in unbiased manner?	6	DR	A comprehensive preliminary EIA (Project Identification File) was prepared in 2010.	OK	OK

9.6	If a stakeholder consultation process is required by regulations/laws in the host country, has the stakeholder consultation process been carried out in accordance with such regulations/laws?	6	DR	N.A. Turkish law on EIAs does not require a stakeholder consultation process for this project type and size.	OK	OK
9.7	Is a summary of the stakeholder comments provided in the VCS PD?	6	DR	Section 6 of the VCS-PD describes that positive feedback was received in informal consultations with local stakeholders. Local residents are impacted by the project in two major ways. First, some private land is expropriated or bought by the project company and local residents are very interested in the expropriation process and the compensation amounts. Second, some local residents are able to find employment during the construction and operation phases of the project. The water in the project area is not used for any irrigation activities so that resource conflicts are limited.	CL15	OK
9.8	Has due account of any stakeholder comments been taken by PPs and reflected in the VCS PD?	6	DR	PPs are following the nationally mandated process in case expropriation of land for project use is required.	CL15	OK
10. Environmental impacts						

10.1 Is the documentation supplied by the PPs regarding environmental impacts relevant and accurately reflected in the VCS PD?	5	DR	The preliminary Environmental Impact Assessment (PIF) and the EIA-Not required Statement by the Ministry of Environment and Forestry were provided to the validation team.	OK	OK
10.2 Is an environmental impact assessment (EIA) required for the VCS project activity? <i>Note: determine by using a review of relevant legislation and local expertise.</i>	5	DR	No. Only a preliminary Environmental Impact Assessment was prepared. The Ministry has stated that the project does not have significant environmental impacts and that no formal EIA is necessary	OK	OK
10.3 In case an EIA is required, has the EIA has been approved by local authorities and is the outcome accurately reflected in the VCS PD?	5	DR	See findings to 10.3.	OK	OK

10.4 Does the VCS PD include a brief description of the environmental effects of the project, including transboundary?	5	DR	In its conclusions on the preliminary Environmental Impact Assessment Ministry has stated that the project does not have significant environmental impacts. The validation team has confirmed during the site visit that the weir will be equipped with a fish ladder and that a minimum flow will be released to the river bed.	CL16	OK
10.5 Are those effects properly addressed in the design of the project activity?	5	DR	The validation team has confirmed during the site visit that the weir will be equipped with a fish ladder and that a minimum flow will be released to the river bed.	CL16	OK
10.6 Does the project comply with environmental legislation in the host country?	5	DR	After reviewing the preliminary EIA, the Provincial Directorate of Environment and Forestry has concluded that a formal EIA is not necessary. There are no further environmental permits necessary.	OK	OK

Table 2: Desk Review Findings

No.	CAR/CLR	Observation (CAR/CL)	Reference	Summary of project owner response	Validation team conclusion
6		CL01. The project description in section 1.9 talks about a dam of 46 m and a weir of 8 m. Please clarify. It was confirmed during the site visit that the project does not contain a dam, but that there are four weirs on the Karakus river and its tributaries.	2.1	The PDD is revised accordingly.	The response is accepted The CL is closed.
7		CL02. Please clarify the length of the 154 kV transmission line.	2.1	The information on the transmission line is omitted from the PDD.	Review 1: It is the understanding of the validation team that the transmission line is part of the project. The length and voltage of the transmission line should therefore be included in the project description.
8		CL03. The PIF and the Feasibility Study talk about 4 diversion weirs, instead of just one. Please clarify in the PDD. It was confirmed during the site visit that the project does not contain a dam, but that there	2.1	The PDD is revised for 4 diversion weir.	Review 1: The response is accepted The CL is closed.

			are four weirs on the Karakus river and its tributaries.			
9	01		CAR01: According to the VCS-PD the installed capacity is 24.45 MW and the power production is 68.74 GWh. The feasibility study talks about 22.82 MW and 68.74 GWh. The license of 01 May 2008 talks about 24.45 MW for Umut II and 124.894 GWh for Umut I-II-III. The turbine purchase contract of 05 June 2008 talks about 3 x 8.15 MW and the PIF of March 2008 talks about 22.82 MW and 81.093 GWh. Please clarify the differences. The project description needs to reflect the latest expected status. The investment analysis needs to reflect the status at the time of decision-making. In table 1 of the VCS-PD power production is given as 82.635 GWh while in section 1.9 it is given as 68.74 GWh. Please clarify which number is used and what the source is. During the site visit it was determined that the investment decision was taken on 05/06/2008, the date when the turbine contract was signed. According to the contract the turbine capacity was 24.45 MW.	2.1	The different figures come from different generation numbers suggested by different sources (i.e. Feasibility Study and Lender's Engineer's report). The Project will have 3 turbines with 8.15 MW each, adding up to 24.45 MW. The Project is expected to generate 68,740 MWh/a, according to the Feasibility Study prepared by Coyne&Bellier and approved by DSI. Therefore this figure is used in the baseline and investment analyses, as it reflects the most realistic scenario and the state of the PP's information at time of investment decision.	The response is accepted The CAR is closed.
10		04	CL04. In section 3.2 of the VCS-PD it is stated that "the project boundary is limited by the National Electricity Grid of Turkey." Please refer to the definition provided in	4.3.1	This formulation and the figure are revised, as requested.	Review 1: The response is accepted

			ACM0002 for a clearer definition. According to ACM0002 “the spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system³ that the CDM project power plant is connected to.” Please also remove the term “project boundary” from Figure 2 in section 1.9 and from the first paragraph of section 1.9.			The CL is closed.
11	02		CAR02: Project emissions from of CH₄ from the reservoir should be considered in Table 8, even though they are demonstrated to be negligible in section 4.2 and properly excluded. For reference see page 6 of ACM0002.	4.3.2	The PDD is revised, as requested.	The response is accepted The CAR is closed.
12		05	CL05: Please provide supporting evidence for the value of APJ=113,696. The footnote refers to technical design documents calculated by the project owner. These should be made available to the validation team during the site visit.	4.3.2	Following the site visit, it has been confirmed that the weirs are sufficiently low and the reservoir sufficiently small so that CH₄ emissions from the reservoir do not have to be calculated.	The response is accepted The CL is closed.
13	03		CAR03: In step 1, please justify the choice of the Turkish National grid as the relevant electricity system. In particular, discuss	4.5.1	- Step 1 revised, as requested. - Weighted average is used and calculations are corrected, as requested. - Step 6 is revised, as	The response is accepted. The CAR is closed.

			that no regional power system is applicable. Please note that the Tool suggest an analysis of transmission constraints and price differences within the electricity system. • In step 4, check that EFgrid, OMsimple, y has been calculated as weighted average of the years 2006-2008. It appears that it has been calculated as straight average. • In step 6, check calculation of UNFCCC Threshold (GWh) in table 17. • Please check the list of BM plants. It appears that the following plant has been erroneously excluded: Borcka HES (2007). • Demonstrate that using the official GHG inventory data fulfils the requirements of the Tool, i.e. Option B for Simple OM and equation (7). Since the top of the formula comes from the GHG inventory and the bottom (EGy) from TEIAS, it needs to be shown that the same power plants are covered. This is particularly important because the strict application of equation (7) with only TEIAS data results in a smaller grid factor.		requested. • The list of BM power plants is revised, as requested. • For the use of national GHG data, an explanation is added to the PDD.	
--	--	--	---	--	--	--

14		06	CL06: - Justify that 2008 is the most recent data available at the time of submission to the DOE. - In order to enhance transparency, please provide data sources for tables 16 and 17 in footnotes. - Please clarify under step 6 that Option A2 has been chosen. - For coal and lignite, please provide supporting evidence that applying the efficiency factor for CFBS technology is conservative. - For coal, please provide supporting evidence that emission factor of 90.8 is conservative. According to IEA statistics (http://www.iea.org/stats/coaldata.asp?COUNTRY_CODE=TR) a mix of coking coal and other bituminous coal is used in coal-fired power plants. - For transparency, please provide the list of VER projects that were excluded from the list of BM plants - Clarify whether autoproducers have been	4.5.1	- The latest data of 2009 is added to the PDD. - Sources for the grid generation data are added to the PDD, as requested. - Explanation about the use of Option A2 is added to the PDD, as requested. - Explanation about the selection of efficiency rate is added to the PDD. - Coal emission factor is revised for other bituminous coal. - A partial list of carbon projects is provided in the spreadsheet. A complete list of voluntary project is unavailable, as there is no central database for such projects that include all projects in Turkey. - In line with the UNFCCC "Tool to calculate the emission factor for an electricity system" Autoproducers are <i>not</i> filtered out from the cohort. Autoproducers are also a part of the OM calculations. They are discussed for common practice analysis. Please note that Autoproducers are grid-	The response is accepted. The CL is closed.
----	--	----	---	-------	---	--

			excluded from the BM cohort as stated in the spreadsheet. This does not appear to have been done consistently According to the Tool the BM cohort includes “the set of power capacity additions in the electricity system that comprise 20% of the system generation (in MWh) and that have been built most recently. If 20% falls on part capacity of a unit, that unit is fully included in the calculation.” The capacity included in the BM calculation significantly exceeds 20%. Please identify the unit on which the 20% falls.		connected power plants that also sell to the grid, but mostly generate for their own purposes. This 20% issue is addressed in the baseline spreadsheet and the PDD, as requested.	
15	04		CAR04: As currently stated scenario 2 in section 2.5 does not qualify as a baseline scenario, since it does not provide the same amount of electricity as the project. It should be stated that the additional electricity will be provided by grid-based electricity <u>from existing and new</u> power plants. Scenario 3 should be amended to “renewable or non-renewable energy project <u>at the project site.</u>”	5.1.2	The scenarios are reformulated accordingly.	Review 1: The response is accepted The CAR is closed.
16	05		CAR05: The period of assessment (time horizon) is 10 years, including 9 years of	5.2.4	The investment horizon is revised to 20 years. A residual value is added	Review 1: The response is accepted

			operation. This is equal to the minimum period suggested in the Guidance, but not appropriate to the project type, which is expected to operate 40-50 years. A 20-year period of assessment should be chosen. No residual value has been included in the final year of the assessment. The residual value should be calculated based on a 40 to 50-year depreciation period.		to the cash flow at the end of investment horizon.	The CAR is closed.
17	06		CAR06: Power prices have been chosen as 6.5 €-cents / kWh, however no supporting evidence has been provided. The use of the feed-in tariff (€55/MWh) is acceptable as the base case in the investment analysis. However, historical wholesale prices should be considered as the upper variant in the sensitivity analysis unless it can be shown that they are not relevant to the decision-making by the company. This could for example be done by providing the original investment analysis that was provided to the Board of the company or a bank providing an investment loan or by demonstrating that the company (or similar companies) is consistently using the feed-in tariffs rather than the whole-sale prices.	5.2.4	The investment analysis has been revised based on the electricity price generated by renewable sources, guaranteed by the Renewable Energy Law of Turkey (€55/MWh). A historical price analysis based on the actual revenues of the PP from its other hydropower plants is provided to the DOE for informational purposes. The sensitivity analysis is revised in the PDD accordingly.	The use of the feed-in tariff (€55/MWh) is acceptable as the base case in the investment analysis. In addition to the feed-in tariff, IPPs have the option to sell power to TEIAS at the market price (System Imbalance Price). Over the past several years there have been many institutional changes affecting the prices at which IPPs can sell to TEIAS. However, even though prices fluctuate greatly on a seasonal as well as on an hourly basis, net prices are roughly

			Please also note that no Residual Balance Adjustment and grid connection fee have been deducted, and no transmission losses have been assumed.			comparable to the feed-in tariffs and within the sensitivity range considered in the PDD. Sources: http://www.invest.gov.tr/en-US/infocenter/publications/Documents/ENERGY.INDUSTRY.PDF http://www.adagunes.com/haberler/Electricity_Generation_Report_%5BTurkey%5D.pdf http://enerjiensitusu.com/piyasa_analizi_karsilastirmali.html Historical price analysis of actual revenues. Also note that seasonal price variation is negative for hydro power plants, because power prices are generally lowest in the Spring when water is plentiful. This is in turn due to the large share of
--	--	--	---	--	--	--

						hydro power plants. The response is accepted The CAR is closed.
18	07		CAR07: Assumption for power generation volume is 82.635GWh. Please clarify where the number comes from. See also CAR01. During the site visit it was determined that the investment decision was taken on 05/06/2008, the date when the turbine contract was signed. According to the contract the turbine capacity was 24.45 MW. This deviates from the feasibility study when the capacity was 22.82 MW. The expected generation amount has to reflect this increase.	5.2.4	The PDD is revised to 24.45 MW and 68,740 MWh/a, according to the Feasibility Study prepared by Coyne&Bellier and approved by DSI.	The response is accepted The CAR is closed.
19	08		CAR08: CAPEX is assumed as 30.741 million €. The stated source is the feasibility study. However, the numbers in the spreadsheet cannot be traced back to the feasibility study.	5.2.4	The Capex is revised to 23,6 million EUR, according to the Feasibility Study.	Review 1: With the higher capacity a higher CAPEX can be expected. CAPEX can be increased proportionally to capacity.

20	09		CAR09: The project could be implemented by an entity other than PPs. Please note the Guidance on Assessment of Investment Analysis: “Benchmark should be based on publicly available data sources which can be clearly validated by the DOE.” (1) Cost of Equity: Risk-free rate of 5% is not substantiated. Stated source for risk premium now states 9.13%, instead of 9.75%. (2) Loan interest rate: No supporting document is provided for rate of 7,25%. (3) It is not clear how the formula for $WACC_{pretax}$ was derived.	5.2.4	- CoE is revised with another source (World Bank report²) suggesting that for HEPPs in Turkey a CoE of 15% is appropriate. Therefore risk free rate or risk premium data are not used any more. - Loan interest rate refers to the cash flow of the PP (submitted to the DOE) at the time of investment decision. - The formula for WACC is added to the PDD.	Review 1: - OK - The loan interest rate of 7.35 is conservative. A potential published source are “real interest rates” from Turkish Central Bank 2008-II Inflation Report, page 59. Spource: http://www.tcmb.gov.tr/yeni/eng/ - OK The response is accepted. The CAR is closed.
21	10		CAR10: OPEX were calculated as 7% of revenues or 376,000 €. This is not conservative compared to the numbers presented in the INEEL study: approx.: USD 250,000 (in 2002 \$). Please demonstrate conservativeness of the numbers.	5.2.4	The PDD is revised with other O&M costs. The O&M figures from the Lender’s Engineer report is used, because it is based on a detailed Section focusing on these costs (this study is submitted to the DOE). Considering that an internationally renown company has calculated these estimated O&M costs for a lending bank for this specific Project, the figures can be seen as	O&M costs of 2,5% of capita costs are based on the Fichtner Report. The response is accepted and the CAR is closed.

² http://www-wds.worldbank.org/external/default/WDSPContentServer/WDSP/IB/2009/05/11/000333037_20090511030724/Rendered/PDF/468080PAD0P112101Official0Use0Only1.pdf

					dependable and of high quality.	
22		07	CL07: Please state the date of the investment decision in section 2.5 of the VCS-PD. During the site visit it was determined that the investment decision was taken on 05/06/2008, the date when the turbine contract was signed.	5.2.4		The response is accepted and the CL is closed.
23		08	CL08: For the Barrier Analysis, please clarify what is the current financing situation of the project. Equity%, loan%, guarantees from mother company for loan, what is the interest rate of the loan? It may be appropriate to address this in the investment analysis. For example, if no loans are used, then equity IRR could be used. Interest rates on past loans or loan offers could be used to substantiate the loan interest rate. The time of the investment decision should also be reviewed. At what time did the PP really decide to go forward (in 2008, after the 23 million were received from Bank A). Is it appropriate to talk about a second investment decision in mid 2010 when mobilisation on site is starting? What happened to enable the project to go forward at that time?	5.3.1	The Project, as of time being, has no debt finance and is being fully financed by equity. Please note that this increases the required rate of return substantially. The estimated E/D ratio was 25/75 at the time of investment decision.	The response is accepted and the CAR is closed.

24		09	CL09: As supporting evidence for the barrier analysis, please provide to the validation team the loan agreement with Bank A and the letter terminating it. Please also provide the letter of guarantee from Bank B and the loan proposal by Bank C.	5.3.2	Barrier analysis is not used.	The response is accepted and the CAR is closed.
25	11		CAR11: The CDM Additionality Tool requires a Common Practice Analysis (Step 4). "Unless the proposed project type has demonstrated to be first-of-its kind (according to Sub-step 3a), the above generic additionality tests shall be complemented with an analysis of the extent to which the proposed project type (e.g. technology or practice) has already diffused in the relevant sector and region."	5.4.1	A common practice analysis is added to the PDD, as requested.	Similar projects are (a) HEPPs (b) Same DSI Region (c) Not autoproducers or government owned (d) Similar capacity (e) Similar load factor (f) Multiple diversion weirs The response is accepted The CAR is closed.
26		10	CL10: Ensure consistency of variable names with ACM0002: $EG_{\text{facility}, y}$ should be used in the Monitoring Plan, not EG_y .	6.1	The PDD is revised accordingly.	The response is accepted The CL is closed.
27		11	CL11: Please identify and provide the relevant national standards that require the periodic maintenance of the electric meters. Are these contained in the Communiqué?	6.10	According to the "Communiqué for Metering Devices to be used in the Electricity Market", Article 9,	The response is accepted The CL is closed.

					paragraph b, the periodic calibration shall be made every 10 years³. The calibration will be carried out by TEIAS. The devices are sealed during first operation of the plant. In case there is a significant inconsistency (assessed by TEIAS) between two devices, the meters will be calibrated by TEIAS before the 10 years period. In such a case, a template ⁴ will be used. The Project owner can also request calibration of the meter(s). The calibration stations are inspected by the central organization of the same Ministry⁵. “Communiqué for Metering Devices to be used in the Electricity Market” is submitted to the DOE.	
28		12	CL12: Please specify the intervals for the periodic calibration of the power meters. Please identify the relevant national standards. Are these contained in the Communiqué?	6.11	Please see CL11.	The response is accepted The CL is closed.
29		13	CL13: According to section 3 of the VCS-PD “The personnel of the Project Owner are	6.16		The response is accepted

³ “Communiqué for Metering Devices to be used in the Electricity Market”, Article 9, paragraph b.

⁴ http://www.teias.gov.tr/mali/GDUY/PRO_FORM/OLCUM/DAG02.xlsa

⁵ Source: “Communiqué for Metering Devices to be used in the Electricity Market”, Article 10

			responsible for the preparation of the monitoring report.” The overall responsibility should be assigned more precisely to one position within the organization, for example the plant manager. This should be included in section 3. During the site visit it was discussed that Ms. Aydin at the Nisan head quarter in Ankara will be responsible.			The CL is closed.
30		14	CL14: Section 1.9 of the VCS-PD claims the following additional environmental benefit. “Treatment of degraded catchments area of the project to control the soil erosion, resulting in regeneration of natural forests and other ecosystems.” Please provide further substantiation or remove the statement.	8.1	This statement is removed from the PDD.	The response is accepted The CL is closed.
31		15	CL15: According to section 6 of the VCS-PD the feedback from local stakeholders was positive. Please describe how typical issues such as expropriation of land, competition for water (e.g. irrigation, drinking water), impact on fish population, and excavation waste have been resolved with local stakeholders.	9.7, 9.8	This statement is removed from the PDD.	The response is accepted The CL is closed.
32		16	CL16: Please provide details on environmental impacts and mitigation efforts, such as fish ladder and minimal flow requirements in section 5 of the VCS-PD.	10.4, 10.5	A general description of environmental impacts is provided in the PDD.	The response is accepted The CL is closed.
33		17	CL17. Please add history of the project in section 1.4., incl. Turbine Purchase Contract	VCS PD 1.4	A chronology is added to the PDD, as requested.	The response is accepted The CL is closed.
34		19	CL19. Please add new contact person for Nisan in Section 1.15. During the site visit it was discussed that Ms. Aydin will fill this role.	VCS-PD 1.15	The PDD is revised, as requested.	The response is accepted The CL is closed.

35		18	CL18. Please clarify whether Mavi is a project participant. If yes, please list Mavi in Table 5.	VCS-PD 1.15	Mavi Consultants is not a project participant, but only acts as carbon consultants for this Project.	The response is accepted The CL is closed.
36		20	CL20. Monitoring of EGy is under the control of TEIAS. During site visit it became clear that for some of the projects (Umut-3) TEIAS does not read the meters on-site but rather reads meters at the nearest substation. As a result, no protocols are printed out and signed by the site manager. In order to avoid problems at verification, please review section 3.2 to ensure that the monitoring plan reflects actual monitoring practice. For items that are not under the control of the PPs, some more flexible language may be chosen. Please clarify in the PDD to what extent EGy is cross-checked against the internal data (from SCADA system) of the PPs and who performs such a cross-check. During the site visit it became clear that for some of the projects (Umut-3) the internal data is sent to Nisan's Ankara office where it is cross-checked against the TEIAS data.	6.2	The PDD is revised, as requested.	The response is accepted The CL is closed.
37		21	CL21. In section 4.2, the table for GENy should clarify in the Description that low cost, must run resources are excluded. In section 4.2, the table for $\eta_{m,y}$ makes reference to Table 25. However table 25 does not include $\eta_{m,y}$.	VCS-PD 4.2	The PDD is revised, as requested.	The response is accepted The CL is closed.

38		22	CL22. Clarify status of EIA for 154 kV power transmission line.	VCS-PD 5	The EIA of the TL is not in the responsibility or under control of the PP. s	The response is accepted The CL is closed.
----	--	----	---	-------------	--	---

Table 3: Forward Action Requests				
No .	FAR	Reference	Summary of project owner response	Validation team conclusion
1	FAR01: Provide revised PIF and EIA Not Required Statement, due to increase in capacity from 22.82 MW to 24.45 MW.			